Supplement to: Channel Estimation for Rydberg Receivers Using Learned Hankel Filtering

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***Abstract*—This supplement explains what the measurements can determine, the cost and limits of the filter, and the tests added in response to review. The original trained models and main-study results are unchanged. New small tests are exploratory: they do not establish a new robustness guarantee or replace the original confidence intervals.**

I. Local Recovery and Reference Calibration

For one array element, let $g, s_p \in \mathbb{C}^K$, with $\mu_p = g^T s_p + b_p$ and noiseless amplitude $a_p = |\mu_p|$. Use real coordinates $x = [\operatorname{Re} g^T, \operatorname{Im} g^T]^T$. When every $\mu_p \neq 0$, differentiation gives

$$da_p = \operatorname{Re}\{\mu_p^* s_p^T dg/|\mu_p|\}, \tag{1}$$

$$A_p = [\operatorname{Re} c_p^T, -\operatorname{Im} c_p^T], \qquad c_p = \mu_p^* s_p/|\mu_p|. \tag{2}$$

Here $A \in \mathbb{R}^{P\times 2K}$ is the real amplitude Jacobian.

Local statement. If $\operatorname{rank} A = 2K$, the noiseless amplitude map is one-to-one in a sufficiently small neighborhood of this g. To see this, choose $2K$ independent rows. Their square Jacobian is invertible, so the inverse-function theorem supplies a local inverse for that measurement subset. The remaining measurements cannot remove this local injectivity. This is not a global uniqueness theorem: distant channels may give the same measurements. A rank-deficient Jacobian also does not, by itself, rule out all forms of nonlinear local injectivity.

For small perturbations at a full-rank point, least-squares inversion has first-order sensitivity bounded by $1/\sigma_{\min}(A)$. Thus $P \geq 2K$ is necessary for regular full-rank local recovery of an unrestricted row. It is not sufficient. Full row rank of S permits the EM linear solve but does not ensure full rank of A. These are standard local arguments in an affine phase-retrieval setting, not a new identifiability theorem [1]. Sparse channel constraints require their own parameterization and Jacobian; a path-count condition is not derived here.

A. Rician Information

For known complex variance $v > 0$ and $a > 0$, the amplitude score is

$$h(z, a, v) = -2a/v + (2z/v)I_1(2za/v)/I_0(2za/v). \tag{3}$$

Define $w(a, v) = \mathbb{E}[h(Z, a, v)^2]$, with expectation under the Rician observation law. For independent pilots the real Fisher matrix is

$$J = A^T \operatorname{diag}\{w(a_p, v)\}_{p=1}^P A. \tag{4}$$

Each magnitude observation contributes an outer product of rank at most one, not a full-rank complex information block. For finite $v > 0$ and nonzero amplitudes these weights are positive. At a zero field the regular amplitude-coordinate derivation is not used; first-order information about the Cartesian mean vanishes. When the assumptions hold, the rank and conditioning of J give a local diagnostic. Neither a small bound nor a well-conditioned matrix at the true channel guarantees that an iterative estimator reaches the correct solution.

B. What a Reference Does and Does Not Fix

With $B = 0$, g and $e^{j\phi}g$ have identical amplitudes for every ϕ. The Jacobian has the phase direction jg in its null space for nonzero g. A nonzero known reference can remove that symmetry, but arbitrary reference and pilot patterns still require a rank check. For example, if $b_p = c^T s_p$ for some fixed c, then $g' = e^{j\phi}(g+c) - c$ produces the same amplitudes. This can occur with a full-row-rank pilot matrix and a nonzero reference. RSR 0 dB in our experiments means a nonzero reference equal in expected power to one user; it is not the absent-reference case.

There is a separate calibration ambiguity. If the estimator assumes $B' = e^{j\phi}B$, then $G' = e^{j\phi}G$ gives

$$|G'S + B'| = |e^{j\phi}(GS + B)| = |GS + B|. \tag{5}$$

The likelihood therefore cannot identify a constant reference-phase error from these observations alone. With error measured in the original reference frame, even this otherwise exact rotated estimate has NMSE $|e^{j\phi} - 1|^2 = 4\sin^2(\phi/2)$. At 10° this is about 0.0304 (-15.2 dB). This is an illustrative exact-fit ambiguity, not a lower bound on every biased estimator. A smaller withheld loss cannot certify correct absolute phase or recover an unknown reference phase. Time-varying phase errors generally cannot be absorbed into one constant channel rotation. A separate static calibration diagnostic estimates a shared positive amplitude multiplier and noise variance under stated assumptions. It cannot remove this unknown-phase ambiguity.

II. Filter Properties and Computation

Let $H = U \operatorname{diag}(\sigma_i) V^H \in \mathbb{C}^{m\times q}$, $m \geq q$, and

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \qquad \tau > 0. \tag{6}$$

Its singular values are $f_\tau(\sigma) = \sigma^3/(\sigma^2 + \tau^2)$. Thus $0 \leq f_\tau(\sigma) \leq \sigma$ and

$$\|F_\tau(H)\|_F \leq \|H\|_F, \qquad \|F_\tau(H) - H\|_F \leq \sqrt{q}\,\tau/2. \tag{7}$$

The second inequality follows because $\sigma - f_\tau(\sigma) = \sigma\tau^2/(\sigma^2 + \tau^2) \leq \tau/2$.

For a channel vector, let d_i count its occurrences in the Hankel matrix and define $\|g\|_D^2 = \sum_i d_i |g_i|^2 = \|\mathcal{H}(g)\|_F^2$. Antidiagonal averaging followed by lifting is the orthogonal

projection onto Hankel matrices in Frobenius norm. Therefore for fixed realized $\eta \in [0,1]$ and $\tau > 0$,

$$g^+ = (1-\eta)g + \eta\mathcal{H}^\dagger F_\tau(\mathcal{H}(g)), \tag{8}$$

$$\|g^+\|_D \le \|g\|_D, \qquad \|g^+ - g\|_D \le \eta\sqrt{q}\,\tau/2. \tag{9}$$

This bounds the size of one correction, not its error relative to the unknown true channel. It does not prove a contraction between two different inputs, particularly when the controller changes η and τ with the input. It also does not prove convergence of the full estimator. For fixed H, $\tau \to 0$ gives $F_\tau(H) \to H$ and $\tau \to \infty$ gives zero. No low-noise limit is asserted because the learned threshold need not vanish with noise.

The matrix-solve shrinker is smooth for $\tau > 0$, including repeated singular values. However, the controller also uses ordered largest and tail singular values. These features need not be differentiable at every singular-value tie. Finite-gradient numerical checks do not establish global differentiability of the whole network.

A. Why the Rician Measurement Update Is an EM Step

Let the unobserved complex field be $Y = GS + B + W$ and write each entry as $Y_{np} = Z_{np}e^{j\Phi_{np}}$. The receiver observes Z, not the phase Φ. For a current predicted field M, the conditional phase density is

$$p(\phi \mid z, M) = \frac{\exp\{\kappa\cos(\phi - \arg M)\}}{2\pi I_0(\kappa)}, \qquad \kappa = 2z|M|/v. \tag{10}$$

This is a von Mises distribution. Its first circular moment gives

$$Q = \mathbb{E}[Y \mid Z, M] = Z\frac{I_1(\kappa)}{I_0(\kappa)}e^{j\arg M}. \tag{11}$$

At $M = 0$, the phase is uniform and $Q = 0$. These are conditional complex fields, not estimates of the mean magnitude. Since $|Y|^2 = Z^2$ when Z is fixed, expanding $\mathbb{E}[\|Y - GS - B\|_F^2 \mid Z, M]$ leaves $\|GS + B - Q\|_F^2$ plus terms independent of the candidate G. Least squares therefore gives $(Q-B)S^H(SS^H)^{-1}$. This derivation uses independent proper complex Gaussian noise and a supplied positive variance; it does not use a strong-reference or high-SNR approximation.

B. Relaxed EM Is Not the Same as the Full Stage

For fixed posterior field Q and supplied variance $v > 0$, the negative EM surrogate, up to constants, is $\Psi(G) = \|GS + B - Q\|_F^2/v$. Its minimizer is G_{EM} when S has full row rank. The normal equations give

$$\Psi(G) - \Psi(G_{\mathrm{EM}}) = \|(G - G_{\mathrm{EM}})S\|_F^2/v. \tag{12}$$

For $G_\gamma = G + \gamma(G_{\mathrm{EM}} - G)$, the surrogate decrease is

$$\Psi(G) - \Psi(G_\gamma) = \gamma(2-\gamma)\|(G - G_{\mathrm{EM}})S\|_F^2/v \ge 0 \tag{13}$$

for $0 \le \gamma \le 2$. Thus even an over-relaxed step within this interval is a generalized EM step for this quadratic M-step, in exact arithmetic. The supplied-model likelihood is nondecreasing. This result does not extend automatically to the subsequent Hankel correction, nor does it guarantee channel-error reduction, global recovery or convergence of the complete learned estimator. The secondary GD core is not covered by this particular argument.

TABLE I
LEADING WORK PER STAGE, EXCLUDING INITIALIZATION

Operation	Leading order
Predicted fields and EM products	$O(NKP)$
EM solve, current implementation	$O(NK^2 + K^3)$
Spectral controller features	$O(Kmq^2)$
Smooth matrix filter	$O(Kmq^2 + Kq^3)$
Antidiagonal averaging	$O(Kmq)$
Small controller	$O(K \cdot 178)$

C. Operation Counts

For $S \in \mathbb{C}^{K\times P}$ and $G \in \mathbb{C}^{N\times K}$, forming SS^H costs $O(K^2P)$. Spectral initialization forms N weighted $(K+1)$-dimensional Gram matrices and eigendecompositions, costing $O(NP(K+1)^2 + N(K+1)^3)$. These are per observation case, not per stage. The inverse-trace noise proxy adds $O(K^3)$ once.

Here $q = \lceil N/2 \rceil$, $m = N - q + 1$. Caching the pilot-Gram factorization could remove repeated K^3 factorizations, but the quoted timings use the existing implementation. The filter avoids differentiating hard rank truncation; it does not remove the SVD needed for controller features. At large N, both dense spectral features and the dense solve scale cubically. A small parameter count is not a small operation count.

For HS-GS, with $R = q$ candidate ranks, $T_s = 25$ selection updates, $T_f = 100$ refit updates and $C = 4$ sweeps, the dominant repeated work is of order

$$(RT_s + T_f)\,[NKP + NK^2 + K^3 + CKmq^2]. \tag{14}$$

Selection uses fewer pilot columns and has additional scoring and initialization work. This expression is a work estimate, not an exact FLOP count or a prediction of the measured speedup. Bessel functions, library choices, memory traffic and batching also matter. At $N = 32$, HS-GS uses 400 selection and 100 refit updates, compared with ten learned stages. Accuracy at those different budgets, not equal iteration count, is the reported comparison.

III. Existing Dimension Tests and Controller Inputs

The original study already contains one-factor-at-a-time tests, with 400 cases per condition and weights fixed. Except for the changed quantity, the reference condition is $N = 32, K = 3, P = 30$, SNR 5 dB and RSR 12 dB. The pilot sweep's smallest value is ten for three users, not six. It does not test the boundary $P = 2K$. The alternative channels use four clusters of ten rays within $\pm 5^\circ$, ten equal-power paths, or ten paths with geometrically decreasing powers. They match base-channel energy and are tested only at SNR 5 dB and RSR 12 dB. These alternatives are simulation populations, not measured scattering distributions.

TABLE II
ALL ORIGINAL MAIN-STUDY COMPARATORS, UNCHANGED

Method	NMSE (dB)	Time (ms)	Severe (%)
EM adaptive (primary)	-14.42	8.24	0.45
EM learned steps only	-10.41	2.98	22.11
EM global filtering	-12.43	5.61	19.19
EM constant-output control	-9.19	5.61	11.42
GD adaptive (secondary)	-14.59	8.00	0.85
GD learned steps only	-10.44	2.73	21.15
GD global filtering	-12.57	5.38	18.37
GD constant-output control	-9.95	5.36	12.36
HS-GS, rank search	-12.74	525.62	14.25
EM, 10 updates	-10.23	4.10	20.64
EM, 100 updates	-10.28	17.01	23.57
Rician GD, 10 updates	-9.31	4.17	17.86
Rician GD, 100 updates	-10.29	17.40	23.50
Rician GD, stopping rule	-10.27	45.37	23.75
Amplitude GD, 10 updates	-8.80	3.29	20.21
Amplitude GD, 100 updates	-9.65	8.87	26.96
Amplitude GD, stopping rule	-9.61	20.93	27.11
Rician GD, hard Hankel	-11.06	47.39	20.93
Linearized least squares	-6.92	1.06	19.86

Accuracy averages seven primary SNR cells. Time is the original batch-one median at SNR 5 dB, RSR 12 dB. These are not the new OptShrink comparison's data or timing sample.

TABLE III
ORIGINAL TRANSFER RESULTS: GAIN OVER HS-GS IN DB

Changed quantity	Gain	Pointwise 95% interval
$N = 8$	-0.106	$[-0.286, 0.077]$
$N = 16$	1.223	$[1.086, 1.361]$
$N = 64$	0.165	$[0.024, 0.307]$
$K = 2$	0.078	$[-0.073, 0.228]$
$K = 4$	1.529	$[1.415, 1.639]$
$P = 10$	2.334	$[2.136, 2.531]$
$P = 20$	1.427	$[1.273, 1.577]$
Noise input -3 dB	1.226	$[1.095, 1.359]$
Noise input $+3$ dB	1.050	$[0.901, 1.193]$

The full controller receives a spectrum summary, residual/noise proxies and size/stage information. A large spectral tail can reflect either noise or real paths; residuals can reflect noise, a wrong channel, or a wrong receiver model. The features provide context but no unambiguous failure detector. No sign constraint makes the threshold monotone in SNR, RSR or path count. The threshold and mixing can compensate for each other, so we inspect the actual relative channel correction as well as both outputs.

A. Original Follow-up Comparisons

The earlier follow-up trains 15 models and evaluates two fresh sets, each with seven SNR settings and two difficult settings of 200 cases. The six primary comparisons adjust for multiple testing within each set. The two new-data adaptive runs use independently generated training data, one fixed training seed per dataset. They are distinct from the three main seeds sharing one dataset and from the new reviewer diagnostics.

The supervised recipe holds architecture, input pilots and training budget fixed but trains on channel targets with mean

TABLE IV
ORIGINAL FOLLOW-UP AVERAGE GAINS IN DB, UNCHANGED

Comparison	Set 1	Set 2
New-data adaptive vs. global	2.252	2.185
New-data adaptive vs. no filtering	4.250	4.191
New-data adaptive vs. HS-GS	1.939	1.837
Original adaptive vs. supervised	1.015	1.012
Original adaptive vs. final-only	0.697	0.694
Fixed-open mix vs. original adaptive	0.064	0.066

linear NMSE. Selecting its checkpoint by withheld likelihood reduces the gap to approximately 0.73 dB; pooling linear errors across SNRs reduces it to about 0.36 dB. It wins in some cases, including two paths. Its transfer from 21 to 30 pilots was less consistent in validation replay. This comparison does not establish a general benefit of withholding labels. The final-only comparison retrains that alternative and its separate timing sample gives 4.65 versus 2.05 ms; those values do not replace the main timing table. Fixing mixing at one retains all 188 stored parameters and worsens two of three seeds on two-path cases, despite its small average gain.

IV. ADDITIONAL REVIEWER DIAGNOSTICS

These checks were specified after the main study and use frozen selected EM models (seeds 9101–9103). No learned weights or original results were changed. Seven SNR cells and two stress cells each contain 32 fresh worlds: 288 distinct worlds. Eight input-mismatch conditions reuse the same 32 SNR 5 dB worlds, giving 544 condition-cases, not 544 independent worlds. Classical and learned methods share the channel, pilots and noise. All tests use $N = 32, K = 3, P = 30$; the main RSR is 12 dB. The extra cells have RSR 0 dB or two paths, at SNR 5 dB.

The small sample, post-review design and fixed checkpoints make these exploratory checks. The intervals resample whole cases with common weights across methods and learned seeds, separately within each cell. They are unadjusted, conditional intervals, not a new confirmatory success test. The original 400-case cells and their intervals remain unchanged.

A. Same Candidates, Different Rank-Selection Loss

HS-GS fits ranks 1–16 on the first 21 pilots using 25 EM updates and four Cadzow sweeps. Exactly the same candidate channels are scored on the last nine pilots using either squared magnitude residuals or Rician negative log-likelihood. Each selected rank is refitted with 100 updates and all 30 pilots. This changes only the selection score, not the fit budget or observation count. A replay against the original residual-ranked solver agreed exactly on a fresh case; an independent likelihood check agrees with SciPy's Rician density up to observation-only constants.

Across the seven SNR cells, Rician scoring improves HS-GS by 0.182 dB, with interval $[0.104, 0.267]$ dB. The frozen learned estimator retains a gain of 1.282 dB, with interval $[1.092, 1.485]$ dB. This score difference explains part, but not all, of the gap in this small check.

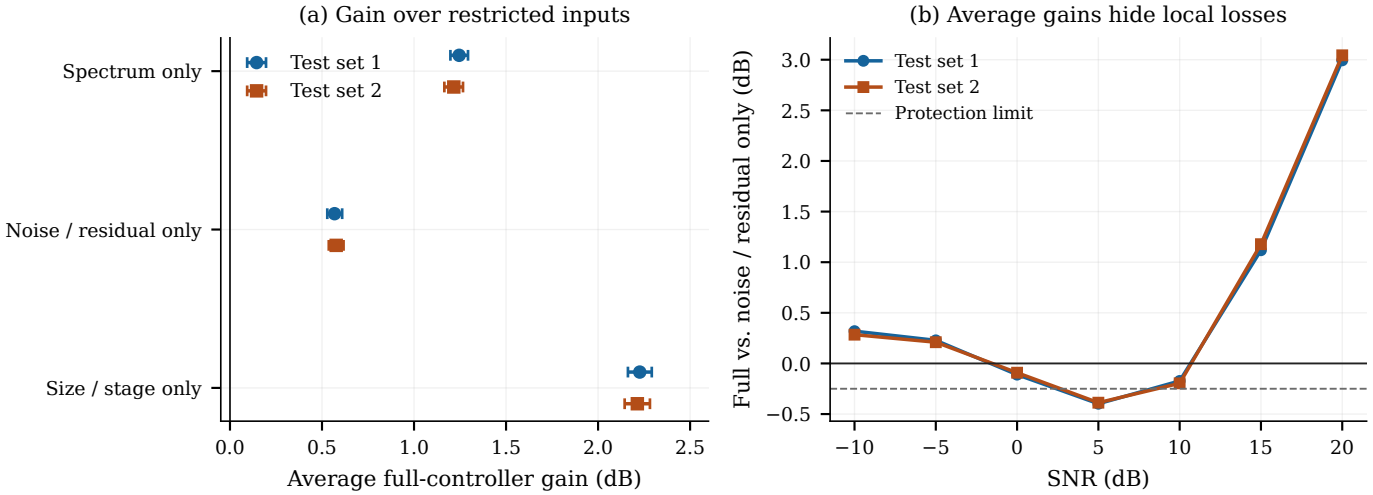


Fig. 1. Original smaller-budget input ablation, unchanged. Full inputs help the average but fail the declared 0.25 dB protection condition. These are separate models trained with 4,096 cases, 12 epochs and three seeds; they are not replacements for the main 16,384-case models.

TABLE V
EXPLORATORY SEVEN-SNR AVERAGES, 32 FRESH CASES PER CELL

Method	NMSE (dB)
EM100, no final filter	-10.162
EM100 + final hard rank six	-11.811
EM100 + final cubic filter	-11.648
EM100 + final OptShrink rank seven	-13.691
HS-GS, residual rank score	-12.778
HS-GS, Rician rank score	-12.961
Frozen learned model, stage-five prefix	-13.353
Frozen learned model, all ten stages	-14.243

B. A Stronger Final-Filter Control

A separate bank of 72 development worlds (eight per SNR/stress cell) selects one global setting for each final-filter method. Selection uses only the withheld Rician likelihood after EM100 on 21 pilots. Hard truncation and OptShrink search ranks 1–15; cubic filtering searches relative thresholds $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$. Every method also has a no-correction option. Candidate counts differ and the grids are small; this is not an exhaustive equal-search-budget comparison. The selected settings are rank six, relative threshold 0.3 and OptShrink rank seven, respectively. The settings are frozen before reading test observations, then used after EM100 with all 30 pilots. None use true channels for tuning.

OptShrink uses the empirical spectral-tail transform in the original algorithm [2], including rectangular zero directions and reduced tail dimensions. A rank equal to full capacity has no tail and is excluded; no correction is a separate candidate. At a tied boundary within a relative squared gap of 10^{-13}, the affected weight is set to its zero-weight limit. Primitive checks cover zero tail, scaling, transpose symmetry, direct rectangular trace evaluation, zero input and the tie rule. These are numerical safeguards, not a new optimality result.

In the matched one-pass test, OptShrink beats the cubic filter by 2.042 dB on the seven-SNR average. Cubic shrinkage is therefore not generally the best structural operation. The complete frozen learned estimator has a 0.552 dB average gain over EM100 plus OptShrink, with conditional interval $[0.432, 0.673]$ dB. It loses 1.040 dB at SNR 20 dB and 2.408 dB at RSR 0 dB, but gains 1.830 dB on two-path cases against this globally selected rank-seven control. That last result does not remove the two-path loss against per-case rank-selected HS-GS.

These comparisons have different update budgets, training/development budgets and filtering schedules. They do not identify the best possible OptShrink-based estimator or reproduce iterative LRHD. No controlled runtime comparison was added; the original 64-fold timing ratio applies only to the original HS-GS implementation.

C. Reference and Noise Inputs

The physical observations stay fixed. For amplitude errors the estimator receives $B_{\text{in}} = aB_{\text{true}}$ with $a = 0.9$ or 1.1; these are amplitude, not power, multipliers. Constant phase errors use $B_{\text{in}} = e^{j\delta}B_{\text{true}}$ with $\delta = \pm 10°$. The ramp uses $B_{\text{in},np} = B_{\text{true},np}e^{j\delta p/(P-1)}$, $p = 0, \ldots, P-1$: the assumed phase error grows from zero to $\pm 10°$. Noise errors supply $v_{\text{in}} = v_{\text{true}}10^{\Delta/10}$, $\Delta = \pm 6$ dB. These are input-mismatch stress tests, not measured hardware drift. They test errors relative to a fixed true reference, not a dynamically changing atomic receiver or a joint calibration algorithm.

Supplying noise variance 6 dB too high worsens the learned estimate by 3.048 dB relative to matched inputs, and makes it 1.409 dB worse than Rician-ranked HS-GS on these paired cases. This is a new observed failure, not evidence of general variance robustness. Amplitude errors also cause clear losses. Constant reference-phase errors rotate every estimated channel by the same phase; replay verifies the equivariance and unchanged phase-aligned errors. Alignment uses truth only as a diagnostic and is not an available inference correction. Absolute-frame errors in the table remain the reported errors.

TABLE VI
INPUT MISMATCH AT SNR 5 DB: LOWER NMSE IS BETTER

Assumed input error	Learned (dB)	HS-Rician (dB)	Learned loss vs. matched (dB)
None	-14.043	-12.496	0.000
B amplitude -10%	-11.613	-9.799	2.430
B amplitude $+10\%$	-12.252	-11.328	1.791
B phase -10°	-11.685	-10.730	2.358
B phase $+10^\circ$	-11.572	-10.564	2.471
Phase ramp $0 \to -10^\circ$	-13.267	-12.029	0.776
Phase ramp $0 \to +10^\circ$	-13.204	-11.774	0.839
Noise variance -6 dB	-13.806	-12.567	0.238
Noise variance $+6$ dB	-10.995	-12.404	3.048

D. Depth and Controller Trajectories

The stage-five output is taken from the unchanged ten-stage forward pass, with its original stage feature $(t+1)/10$. It is not a five-stage model retrained with $(t+1)/5$. Completing ten stages improves the seven-SNR average by 0.890 dB, with interval $[0.815, 0.964]$ dB; it does not help every cell. This prefix experiment does not determine an optimal depth or answer whether a trained twenty-stage model is better.

At the ordinary SNR 5 dB condition the mean relative threshold falls from about 0.915 at stage one to 0.362 at stage ten. Mean mixing is about 0.99996. Thus the useful adaptation is primarily in threshold strength, not abstention. Large relative corrections in a failure condition do not prove which feature or prior caused the failure.

V. CONTROLLED IN-LOOP OPTSHRINK COMPARISON

This experiment uses the frozen original three EM models. A separate development set has 32 worlds at each of seven SNRs. For both in-loop and final-only OptShrink, the candidate grid includes ranks 1–15, mixing 0.25, 0.5 or 1, update budgets 10, 20, 50 or 100, and no filtering. Candidates fit 21 pilots; the other nine score Rician likelihood. Noise-dependent settings are frozen before generating 128 fresh worlds at each SNR and at each of two stress settings, giving 1,152 test worlds. All test methods use all 30 pilots. Reference strength is 12 dB and channels have three to seven paths, except in the labelled stress cells.

Unlike the first final-filter diagnostic, this control shrinks the lifted channel after each EM update and averages its antidiagonals. It is an OptShrink-based Rician reconstruction, not full LRHD on directly observed noisy Hankel matrices. Hankel lifting and nonlinear reconstruction produce dependent errors, so matrix-denoising optimality guarantees are not claimed. Both noise-dependent and global settings are retained in the results.

The selected in-loop control uses ten iterations up to SNR 10 dB and twenty at 15 and 20 dB. The learned model loses 0.143 dB on the declared seven-SNR average. It gains 0.723 dB at -10 dB but loses 0.796 dB at 20 dB. Its severe-error rate is 0.89%, versus 5.02% for the selected in-loop method. At -10 dB these rates are 6.25% and 35.16%; the other primary SNR cells have none in this sample. The lower mean per-case linear error of the learned method is a different metric, not a reason to replace the original primary metric after seeing the results.

At RSR 0 dB and SNR 5 dB, the selected in-loop control has NMSE -5.026 dB, versus -7.069 dB for learned and -8.307 dB for HS-Rician. However, its 50- and 100-iteration controls reach -8.790 and -8.981 dB. The apparent learned advantage over the selected control therefore does not establish superiority over the in-loop family at weak references. For two paths, HS-Rician reaches -17.833 dB, learned -16.814 dB and selected in-loop -16.407 dB. No algorithm wins every comparison.

End-to-end single-thread CPU timing at SNR 5 dB includes initialization, noise-level lookup and the selected reconstruction. Mean times are 5.67 ms for one learned model, averaged across three trained models, and 6.20 ms for selected in-loop OptShrink. Final-only filtering takes 3.99 ms and HS-Rician 356.65 ms in this timing sample. These are not ensemble inference times, embedded-device measurements or GPU benchmarks. The original 64-fold ratio applies only to the earlier HS-GS comparison. Saved outputs were recalculated; independent OptShrink primitive replays agree within 5×10^{-15} and selected learned replays agree exactly.

VI. STATIC NOISE AND REFERENCE ESTIMATION

This exploratory check estimates a single positive reference-amplitude multiplier a shared by the array and a shared complex noise variance v. Reference shape and phase remain supplied. It does not estimate per-antenna drift or choose new measurements. With posterior field Q, let $R = I - S^H(SS^H)^{-1}S$. One conditional least-squares update is

$$a^+ = \mathrm{clip}_{[0.25,4]} \frac{\mathrm{Re}\langle BR, QR\rangle_F}{\|BR\|_F^2}, \tag{15}$$

$$G^+ = (Q - a^+B)S^H(SS^H)^{-1}, \tag{16}$$

$$v^+ = \mathrm{clip}_{[10^{-6},10^3]}\, \mathrm{mean}\{Z^2 + |M^+|^2 - 2\,\mathrm{Re}[(M^+)^*Q]\}, \tag{17}$$

where $M^+ = G^+S + a^+B$. Bounds use the normalized simulation units and are algorithm assumptions, not measured receiver limits. If $BR = 0$, the gain is confounded with the channel and the joint fit is rejected. This denominator check is not a global identifiability proof for the full nonlinear problem.

Three choices—unchanged inputs, noise estimation, and joint gain/noise estimation—are fitted on 21 pilots for 5, 20 or 100 iterations. The other nine choose a setting, which is refitted on all 30 observations. The selection loss includes $\log v$:

$$\ell_{\mathrm{nuisance}} = (z - |\mu|)^2/v - \log I_{0e}(2z|\mu|/v) + \log v. \tag{18}$$

Only $-\log(2z)$ may be omitted when comparing different variances. All nine candidates and the final refit count toward processing cost. The calibration preprocessing averages 45.05 ms per case in this CPU check, before the unchanged learned estimator runs. It is not a free robustness improvement. An EM control with the same held-out stopping choice separates stopping effects from nuisance estimation. Corrected inputs are also passed to the unchanged learned model; there is no retraining on these cases.

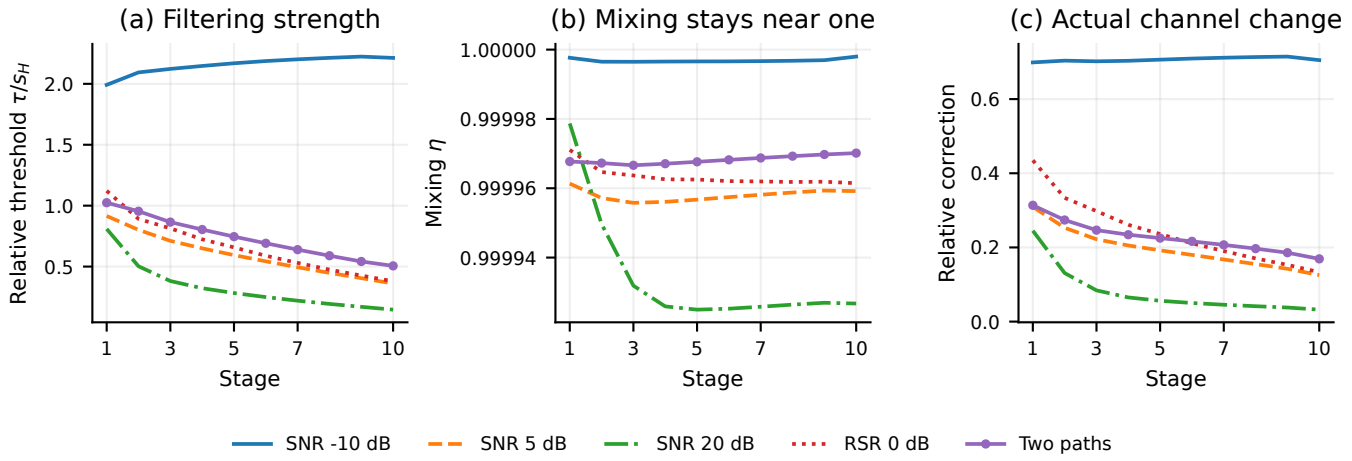


Fig. 2. New frozen-model trajectories, averaged over 32 cases, three model seeds and users. Main SNR curves use RSR 12 dB; weak-reference and two-path curves use SNR 5 dB. All other settings are unchanged. Mixing stays near one, while threshold and actual correction vary. Curves describe outputs; they do not establish causal feature importance or a failure detector.

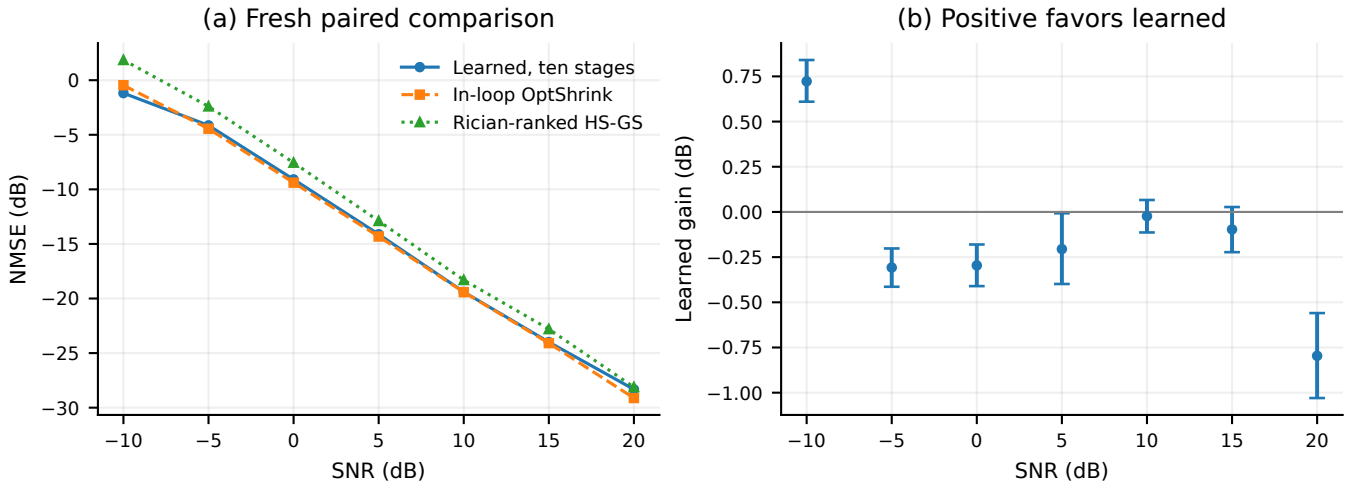


Fig. 3. New in-loop comparison, separate from the original main study. Each SNR has 128 paired worlds. Right: learned gain over in-loop OptShrink; positive means lower error. Intervals are conditional paired-case 95% intervals, with no per-SNR multiplicity adjustment.

There are 32 worlds per SNR/reference cell, at SNR $-5, 5, 15$ dB and RSR 0 or 12 dB: 192 independent worlds. Seven assumed-input conditions reuse them, giving 1,344 condition-cases, not independent worlds. The conditions are correct inputs, variance errors of ± 6 dB, reference-amplitude errors of $\pm 10\%$, joint $+10\%/+6$ dB error, and constant phase error $+10^\circ$.

Average improvement does not make the correction safe in every cell. With weak reference at SNR -5 dB and a $+6$ dB variance error, it worsens learned NMSE by 1.048 dB. At SNR 15 dB with strong reference, a $+10\%$ amplitude error is largely corrected: NMSE changes from -16.346 to -24.107 dB; the true-input diagnostic is -24.253 dB. These are simulation input errors, not observed hardware behavior. Unknown constant phase remains unresolved because channel and reference can rotate together. Likelihood improvement cannot certify absolute phase or reduced channel error. The original known-calibration estimator remains the primary method; this check is not a universal calibration remedy.

TABLE VII
STATIC CALIBRATION: GAIN OF CORRECTED OVER SUPPLIED INPUTS

Input condition	Mean gain (dB)	Worst cell gain (dB)
Correct inputs	-0.013	-0.054
Variance -6 dB	0.832	0.138
Variance $+6$ dB	0.997	-1.048
Amplitude -10%	1.863	-0.072
Amplitude $+10\%$	1.449	-0.413
Joint $+10\%/+6$ dB	2.872	-0.746
Phase $+10^\circ$	-0.008	-0.048

Equal average over six SNR/RSR cells. Positive gain means lower error. These are descriptive point estimates, not a simultaneous robustness guarantee.

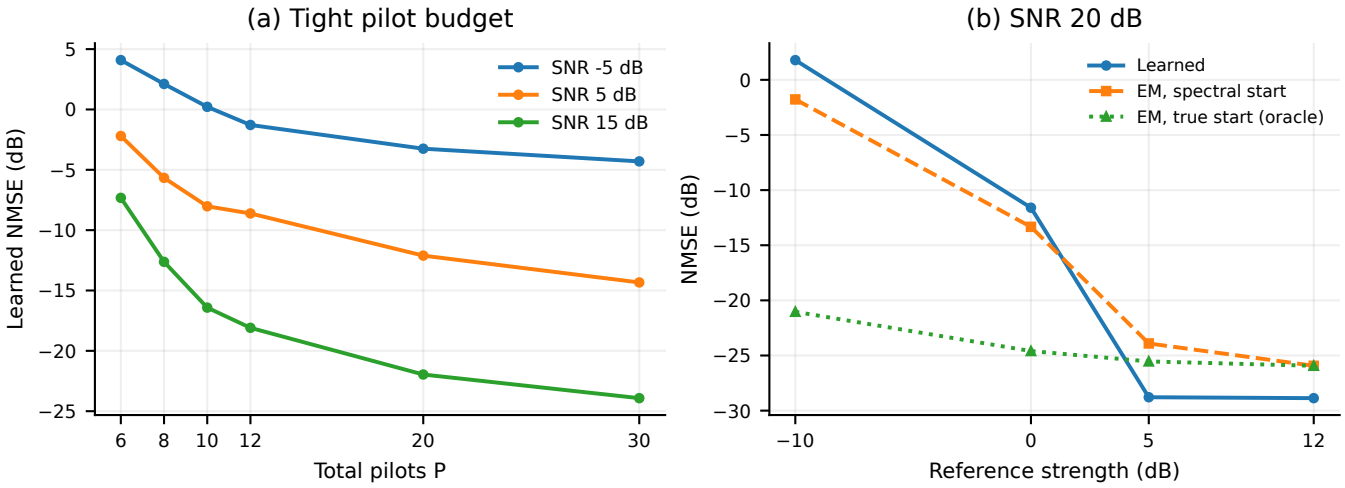


Fig. 4. Exploratory checks, 32 worlds per cell. Left: frozen learned model under changing pilot count. Right: EM initialization diagnostic with the same channel, pilots and noise across reference levels. The true-start curve uses unavailable ground truth and is not a deployable estimator.

VII. Tight Pilot Counts and Weak References

The new pilot sweep has $P = 6, 8, 10, 12, 20, 30$, $K = 3$, SNR $-5, 5, 15$ dB, RSR 12 dB and 32 fresh worlds per cell, or 576 worlds. Learned weights and OptShrink settings are transferred from $P = 30$, without tuning on these tests. HS-Rician fits ranks on $\mathrm{round}(0.7P)$ pilots, scores the remainder, then refits all P. Its rank-fitting subset can be below $2K$ even when P reaches it. Structural coupling across array elements means the unrestricted single-row count is not, by itself, an identifiability theorem for the structurally constrained model.

At SNR 5 dB, learned NMSE is -2.197 dB with six pilots and -14.332 dB with thirty. At SNR -5 dB and six pilots it is 4.089 dB, showing substantial failure. All sampled nonzero-reference full-data row Jacobians have rank six. Nevertheless, at SNR 5 dB the median smallest singular value is 0.172 with six pilots and 2.488 with thirty. Local full rank is not enough for stable finite-noise estimation or global recovery. The cells use different independent worlds; the method comparisons within each cell are paired.

A second check shares the same channel, pilots and complex noise across absent reference and RSR $-10, 0, 5, 12$ dB, at SNR 5, 20 and 60 dB. There are 32 independent worlds per SNR and 480 condition-cases. The SNR 5 dB reference check shares a generation seed and channels with one tight-pilot cell. The two diagnostic phases are not independent replications of each other and are not pooled for uncertainty estimates. At zero reference, sampled Jacobians have rank five and absolute phase is ambiguous. At nonzero reference they have rank six in this sample. At SNR 20 dB and RSR -10 dB, ordinary EM gives -1.773 dB NMSE from spectral initialization but -21.036 dB when initialized at truth. The latter is an oracle diagnostic only. The gap supports an initialization or optimization difficulty; it does not prove global identifiability or exclude controller problems.

A. A Nearly Noiseless Failure

SNR 60 dB lies far outside the training range. Even at RSR 12 dB the frozen learned model gives -6.647 dB, while EM100 gives -65.715 dB. A post-hoc intervention on the same 32 cases disables only learned filtering and retains the learned step sizes; its NMSE is -44.020 dB. Starting the full learned model at the true channel still gives -6.653 dB. Thus filtering can spoil an already accurate estimate in this setting; the failure is not explained solely by a bad initializer. Independent NumPy arithmetic for the no-filter intervention agrees with the PyTorch result within 5×10^{-15}. This is an out-of-range failure and a post-hoc diagnostic, not an independently confirmed switching policy. No low-noise consistency or generally safe filtering claim is made.

VIII. Separately Trained Depth, Width and Training Controls

This reduced-budget study answers the depth and capacity questions using new training, not a prefix of the original network. Each dataset contains 4,096 training and 448 independent validation worlds. Training uses 12 epochs, batch size 32 and Adam with learning rate 0.001. There are seven SNRs from -10 to 20 dB, RSR 0, 5, 12 and 20 dB, and either random three-to-seven-path or fixed two-path channels. This gives 56 conditions, with $N = 32, K = 3, P = 30$. Unlike the original training mixture, this one includes RSR 0 dB. Its scores must not be pooled with the original study.

Seven variants share dataset zero and three initialization seeds: the ten-stage, width-16 baseline; depths five and twenty; widths eight and 32; two experts; and curriculum training. The baseline also trains on two additional independent datasets with two initialization seeds each, giving 25 jobs and 300 completed epochs. Initialization seeds do not create independent training populations. Training has no channel targets. Validation selects the checkpoint using the equal-condition average of withheld Rician likelihood, with a fixed 21/9 pilot

split. The final epoch is also evaluated, not substituted when its test error is better.

Ordinary variants share example order and pilot masks at each epoch. Depth changes the number of learned steps and uses $(t+1)/T$ as its stage input. The two-expert model combines two width-16 controller outputs with a learned softmax gate driven by the three spectral features. It has 374 parameters, compared with 364 for width 32. The experts start at relative thresholds 0.05 and 0.2 rather than the plain controller's 0.1. They do not recover distinct physical channel components. Architecture and initialization both change; any gain cannot be attributed to gating alone.

The curriculum presents RSR 12/20 examples twice per epoch for six epochs, then RSR 0/5 examples twice per epoch for six epochs. Total exposure per example matches ordinary training. The repeated visits share that epoch's pilot mask, so the number of distinct masks per example differs. This is not an isolated test of ordering alone. Neither channel truth nor test errors determine this schedule.

After all checkpoints were fixed, 64 fresh worlds per condition were generated, giving 3,584 paired test worlds. All 25 selected and 25 final models use the same observations, together with EM100 and the previously frozen in-loop OptShrink rule. That rule transfers from its earlier development mixture; it is not retuned on these new tests. A documented test-seed amendment was made before generation to avoid overlap with diagnostic seeds. Training and selection were unchanged.

A. Accuracy and Cost

The reported macro-average gives each condition and initialization equal weight in dB. Five stages are faster, but lose 1.977 dB against ten. Twenty gain 0.181 dB while increasing mean time from 8.04 to 15.60 ms. Their severe-error rate is also higher: 13.46% versus 10.28% in this 56-condition mixture. The depth comparison is therefore not a simple "more is better" result. Twenty stages improve the RSR 0 dB average by 1.547 dB but worsen one low-SNR cell by 1.586 dB. The absolute severe-error rates use a broader, weaker-reference mixture than the original experiment and are not evidence of a change in its saved results.

Width eight loses 0.731 dB and has almost unchanged processing time: matrix operations, not controller size, dominate this implementation. Width 32 gains 0.065 dB on average and reduces severe errors, but loses 0.970 dB in its worst cell. The two-expert gain is only 0.023 dB overall and 0.179 dB for two-path cells; width 32 gains 0.286 dB on those cells. The experts therefore do not demonstrate a useful advantage over the similarly sized plain controller in this check.

The curriculum gains 0.206 dB overall and 0.255 dB at RSR 0 dB, but loses 2.353 dB at SNR 20 dB, RSR 5 dB and two paths. It does not solve weak-reference or few-path estimation generally. No tested variant improves every cell. The original primary estimator is retained; these test results are not used to choose a replacement.

Timing uses one CPU thread, double precision and batch size one, includes initialization, and repeats 32 shared SNR 5 dB/RSR 12 dB worlds three times in shuffled model order. Means average individual-model inference, not ensemble inference. This timing batch is separate from the original and OptShrink timing batches; cross-batch speed ratios are not claimed. The 25-job training invocation took about 103 minutes on the local CPU, including data preparation and validation, but excluding fresh testing.

B. Variability and Scope

The three independent training datasets give baseline two-initialization macro-averages of -12.450, -12.124 and -12.229 dB. Their individual initialization scores are $(-12.391, -12.509)$, $(-11.824, -12.425)$ and $(-12.143, -12.315)$ dB. These few replicates expose variability but do not estimate broad population uncertainty reliably. All use the same test worlds; repeated predictions are not independent channel realizations.

The paired intervals resample 64 whole cases within each condition 2,000 times, retaining paired methods and averaging the three fixed-model scores. They are exploratory, conditional and unadjusted across six comparisons. Small intervals do not remove uncertainty from training or prove an architecture optimum. Most selected checkpoints are the final epoch; 12 epochs do not establish convergence.

The transferred OptShrink rule gives -12.150 dB and 12.11% severe errors; EM100 gives -8.784 dB and 25.53%. The new baseline's -12.390 dB score cannot reverse the earlier controlled OptShrink result: both the training mixture and test weighting differ, and the classical settings were not developed for this mixture. These are transfer controls, not a new claim of tuned-classical superiority.

C. Saved-Result Audit

All 25 histories, 300 epochs, 50 checkpoint hashes and 2,912 prediction files were checked. The latter comprise 1,400 selected-model, 1,400 final-model and 112 classical files. Errors, severe-error rates and headline averages were recalculated from predictions and truth. Matching orders and masks were checked with the stated curriculum exception. One world per test condition was regenerated from its seed; selected and final models were replayed individually on a saved case. The largest replay difference was below 1.2×10^{-15}. Seventeen focused tests cover the new numerical/training/seed checks and OptShrink implementation. The release retains these audit tools and saved observations, not just summary scores. This checks the reported computation; it is not external replication or real-receiver validation.

IX. Related Structural and Sparse Methods

OptShrink estimates singular-value weights from the observed noise spectrum [2]. LRHD combines that principle with repeated Hankel projection [3]; neither our first one-pass diagnostic nor the new EM/OptShrink adaptation reproduces full LRHD. In both cases the starting problem is noisy matrix denoising, not recovery from affine magnitudes. The noise on a Hankel matrix formed from an iterated channel estimate need not meet asymptotic independent-entry assumptions. Neither those guarantees nor universal optimality are asserted here.

TABLE VIII
SEPARATELY TRAINED CONTROLS: ALL 56 CONDITIONS, EQUAL-WEIGHT DB AVERAGE

Variant	Parameters	Selected NMSE (dB)	Final NMSE (dB)	Gain (dB)	Worst-cell gain (dB)	Severe (%)	Time (ms)
Ten stages / width 16	188	-12.390	-12.297	0.000	0.000	10.28	8.04
Five stages	183	-10.413	-10.413	-1.977	-9.540	6.48	4.25
Twenty stages	198	-12.571	-12.571	0.181	-1.586	13.46	15.60
Width 8	100	-11.658	-11.658	-0.731	-3.070	14.14	8.02
Width 32	364	-12.454	-12.454	0.065	-0.970	4.91	8.04
Two experts	374	-12.413	-12.385	0.023	-0.931	8.31	8.55
Curriculum	188	-12.596	-12.634	0.206	-2.353	6.84	8.03

All rows use dataset zero, three initialization seeds, 4,096 training cases and 12 epochs. Gain is relative to ten stages/width 16; positive favors the variant. Severe rates and gains use selected checkpoints. Time is individual-model mean inference time, not an ensemble time. This protocol differs from the original study.

TABLE IX
INITIALIZATION VARIABILITY AND CONDITIONAL GAINS IN THE REDUCED-BUDGET STUDY

Variant	Selected seed scores (dB)	Selected epochs	Gain interval (dB)	RSR 0 gain	Two-path gain
Ten stages / width 16	-12.391, -12.509, -12.269	12, 12, 11	—	0.000	0.000
Five stages	-10.779, -10.906, -9.554	12, 12, 12	$[-2.137, -1.849]$	-1.500	-1.943
Twenty stages	-12.382, -12.579, -12.751	12, 12, 12	$[0.062, 0.401]$	1.547	0.075
Width 8	-11.906, -11.288, -11.781	12, 12, 12	$[-0.779, -0.710]$	-0.158	-0.696
Width 32	-12.546, -12.659, -12.158	12, 12, 12	$[0.050, 0.087]$	0.115	0.286
Two experts	-12.532, -12.478, -12.228	11, 12, 12	$[0.014, 0.035]$	0.100	0.179
Curriculum	-12.676, -12.903, -12.209	11, 12, 12	$[0.153, 0.292]$	0.255	0.117

Initialization seeds are 9201, 9202 and 9203 in order. Intervals are unadjusted conditional 95% paired-case intervals, not intervals over training datasets or initialization seeds. Subgroup gains average all matching conditions; they do not imply improvement in every matching cell.

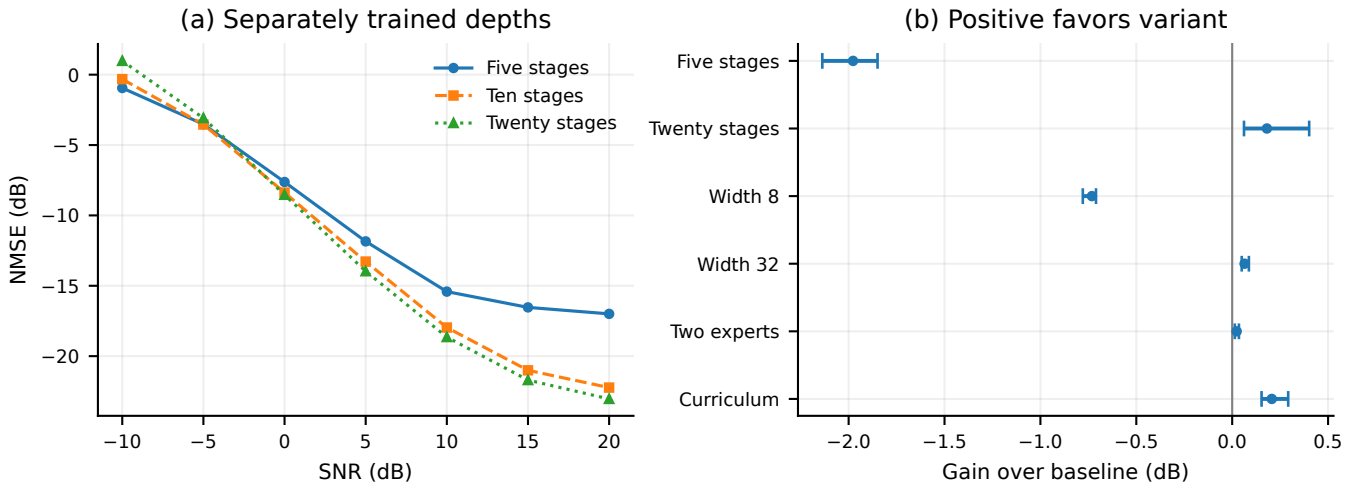


Fig. 5. New reduced-budget training study. Left: selected-model NMSE, averaging four reference levels, two path settings and three initializations at each SNR. Right: gain over ten stages/width 16 across all 56 conditions. Bars are unadjusted conditional 95% paired-case intervals; they exclude training-data and initialization uncertainty.

Structured alternating least squares reconstructs low-rank matrices from linear measurements, including Hankel structure [4]. HSGD addresses partially observed Hankel matrices with sparse gross corruptions [5]. Their losses, observations and guarantees cannot be transferred unchanged to our nonlinear Rician model. They remain important alternative constructions, not irrelevant methods.

Noncoherent compressive channel estimation combines phase retrieval and sparse reconstruction using a designed sensing cascade [6]. GP-bandit beam training uses adaptive near-field beams, Rician-debiased pseudo-amplitudes and sparse phase retrieval [7]; it has no known additive reference. Partially coherent matching pursuit retains complex observations with a shared unknown phase within each packet [8]. That information is absent from our magnitude-only inputs. A fair benchmark must adapt these methods to the same observation and pilot budget rather than importing their published scores.

The multi-shot Rydberg study [9] concerns signal-presence detection with Rician shots. It motivates accounting for obser-

vation statistics and shot budgets, but supplies neither channel-estimation results nor a heavy-tailed noise model for this experiment.

X. What Is Still Not Tested

Depth. The new 5/10/20-stage comparison uses separately trained models, unlike the original prefix diagnostic. It has a fixed small training budget, not an exhaustive search or a convergence test. An optimal depth for larger datasets or other channel populations is not known.

Noise and hardware. The original model has independent complex Gaussian noise before magnitude extraction. Incorrect variance inputs do not test non-Gaussian noise. Finite shot count alone does not imply heavy-tailed noise, and no cited detection model establishes that assumption for this receiver. Optical readout, atomic dynamics and hardware data remain untested. A Rician loss is not automatically matched to an alternative optical or multi-shot measurement statistic.

Structure. Cubic shrinkage is one design choice, not a proved optimal denoiser. The new in-loop comparison covers one OptShrink-based reconstruction. Full LRHD, structured ALS and robust Hankel completion would require their own adapted measurement models and independently selected settings. Existing hard, global, adaptive and no-filtering controls do not exhaust these choices. An angle-domain sparse phase-retrieval method is also a relevant missing comparison, particularly for two-path channels. No superiority over those methods, or over published URformer, is claimed.

Training. Two-path and RSR 5 dB cases already occur in the original training; RSR 0 dB does not. The new smaller study includes RSR 0 dB, and its curriculum has mixed results. No safety-critical protection guarantee follows from average gains. The original pointwise intervals condition on fixed models and shared training data; the fresh-data follow-up is limited and does not estimate broad training-population uncertainty.

Array geometry. A two-dimensional array may suggest multilevel or block-Hankel structure, but its lifting dimensions, normalization, computational cost and training transfer must be defined anew. No 2D array result is reported.

XI. Initialization and Numerical Record

For array element n, form $C_n \in \mathbb{C}^{P\times(K+1)}$ with rows $[s_p^T, b_{np}]$. Let u_n be the dominant eigenvector of $C_n^H \operatorname{diag}(z_n) C_n$ and set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{19}$$

The first K components of $c_n u_n \exp[-j \arg(u_{n,K+1})]$ initialize the channel row. Only the 21 input pilots enter this calculation during training. The withheld columns do not enter the initial estimate.

The original 43 focused tests cover likelihood and derivative checks, noise scaling, finite gradients, repeated singular values, pilot-split isolation, single/double precision agreement and deterministic recovery. The submission audit recalculated 1,120 main, 702 follow-up and 432 input-study result files. Three primary checkpoints replayed eight saved observations exactly; 24 input-study checkpoint replays were also checked. This is not a replay of every historical experiment or an independent hardware validation.

One finite-input SVD failure occurred in the original classical HS-GS test at $N = 64$. The same matrix was recomputed with SciPy's alternative SVD driver; relative reconstruction error was below 3×10^{-15}. The trial was retained with unchanged inputs and iteration budget. Original software versions were Python 3.12, NumPy 2.4.6, SciPy 1.17.1 and PyTorch 2.13.0. The expanded release includes saved inputs, truth for evaluation, predictions, selected/final checkpoints, training histories, settings, source hashes and figure scripts for the reported studies. The separate Overleaf archive contains manuscript files, not experimental data. No GPU agreement or hardware measurement is claimed.

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